

1. A top sprinter reaches 11.2 m/s within 3.15 s of the start of a race. Calculate the average acceleration.
2. Calculate the final velocity of a car initially travelling at 10.0 m/s that accelerates at 1.4 m/s² for 8.0 s.
3. How long does it take for a F1 race car to accelerate at 8.70 m/s² from 150 km/h to 290 km/h as it enters a straight?

4. A truck moving at 36 m/s is decelerated uniformly at 1.5 m/s^2 . What is its final velocity after 5.0 s?
5. A car of mass 1800 kg moving at 90 km/h applies its brakes, exerting a force of 2300 N for 3.5 s. What is the final velocity of the car?
6. A car of mass 1530 kg is travelling at 70.0 km/h along Hepburn Avenue when the driver has to brake sharply to avoid a small child who suddenly appeared on a bike crossing the road.
- (a) If it took 3.5 s to stop the car, calculate the average force exerted by the brakes.

- (b) The driver of the car was an engineer and had a hard-hat sitting on the back shelf. Describe what happened to the hard-hat in this situation and explain why it took place.

7. (a) Explain the difference between **mass** and **weight**. Give an example to help your explanation.

- (b) "Space is great! You are weightless and it is easy to move stuff around." Comment on the accuracy of this statement.

- (c) The acceleration due to gravity on Neptune is 11.1 m/s^2 . Calculate the weight of a 200 kg object on Jupiter.

8. A 70.0 kg cyclist is applying a force of 150 N forwards but is working against a wind producing a friction force of 80 N opposing the movement.
- (a) Determine the resultant force for this situation.
 - (b) Calculate the acceleration of the cyclist.
 - (c) It is possible for the rider to move at constant velocity. In terms of the forces acting, explain how this is possible.
9. You are sitting on a seat at the moment. Explain why you don't fall through the chair and are supported by it.

10. Steven enjoys BASE jumping off cliffs. He particularly enjoys a high cliff in Pakistan called Great Trango, which is 1340 m high.

(a) What is Steven's theoretical velocity after falling for 14.0 s, which is a fall of 1000 m?

(b) Why is your answer to part (a) theoretical?

(c) Steven, who has a mass of 85 kg, is actually moving at 45.0 m/s at this point when he releases his parachute. What force does it exert on Steven if he decelerates to 25.0 m/s in 1.4 s?